# Replication: A Longitudinal Dataset to Characterize IETF RFC Publication and Deployment


## ABSTRACT

Protocol standardization plays an essential role in the successful operation of the Internet and its inherent properties such as reliability, performance and security. However, RFCs and protocols have varying levels of adoption, and identifying factors that relate to protocol adoption success is key to improving the standardization process and ultimately enabling the Internet properties that the community considers relevant. In addition, monitoring the features of RFC development with more predictive power of success, their evolution over time and their representativeness across RFCs allows us to evaluate the overall health of the standardization process.

Recognizing the relevance of the protocol standardization process, in 2021 McQuistin *et al.* [1] characterized the work of the Internet Engineering Task Force (IETF) in Internet standard development by creating and analyzing a corpus of more than 8000 RFCs and related email interactions among authors. They find that the complexity of the standardization process and the time for RFCs to get published have increased over time. Additionally, based on experts' RFC implementation success evaluation from [2], they build a statistical model to identify factors that impact RFC implementation.

Although in [1] the authors made the data available at publication time, many data cleaning tasks to sanitize the metadata that need to be manually updated as new RFC may have new authors, affiliation and geographic representation, and the source for citation count metadata of RFC was shut down in 2022. In addition, the expert evaluation of 155 RFCs from [2] that the authors use to explain factors of success are becoming increasingly dated.

In this paper, we will replicate the work of [1] by creating a new RFC corpus with documents, metadata and features, created using an automated pipeline that can be run repeatedly to include new RFCs, as shown in Figure 1. We will use and evaluate LLMs to replace all manual data cleaning and sanitation tasks and we will train LLM models on the experts evaluation of [2] to extend it to newer RFCs. Figure 2 depicts the yearly count of RFCs in our latest dataset, including more than 700 RFCs published after 2021. This allows us to interpret post-2021 RFC publication trends in the context of the original study [1], replicate all analysis done in [1], and explore the discrepancies. Using our pipeline, we plan to build a monthly corpus of RFCs, metadata and features, and make it publicly available to the community to support further research on the IETF and standardization processes.


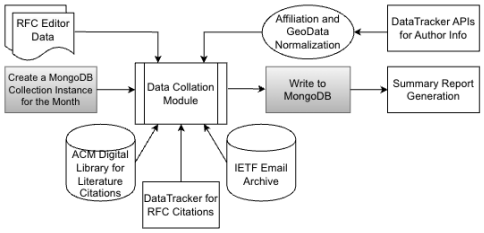


**Figure 1: Diagram of datasets used to create the RFC and standardization process dataset**

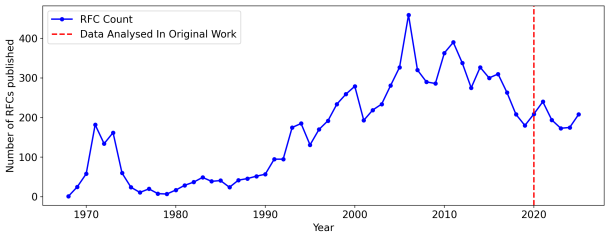


**Figure 2: Count of published RFCs per year until end of 2025**

We plan to develop an LLM-based approach to replace the manual data cleaning and expert annotation steps that currently limit the predictive model of RFC success. The predictive model in [1] relies on manual sanitization of metadata as well as expert evaluation of RFC deployment from the work of Nikkhah *et al.* [2]. In [2], the authors evaluate 251 RFCs, but only 155 of those are used in the final RFC corpus of [1]; as such, the predictive model is effectively based on 155 RFCs out of the more than 8000 RFCs available online, and this expert-labeled set becomes increasingly dated as new RFCs are published. We plan to use LLMs trained on the 251 RFCs and expert annotations from [2] to reproduce the expert labels and extend these annotations to newer RFCs at scale. We will evaluate how well the LLM reproduces the expert annotations by comparing its predictions to the labels in [2], using validation methodologies such as k-fold cross-validation, and reporting standard evaluation metrics along with a brief error analysis of common failure cases. With our proposed pipeline, we would provide a continuously updated RFC corpus, coupled with scalable LLM-based cleaning and annotation, can enable new research that improves our understanding of the standardization process and supports a healthy standard ecosystem.

## REFERENCES


[1] Stephen McQuistin, Mladen Karan, Prashant Khare, Colin Perkins, Gareth Tyson, Matthew Purver, Patrick Healey, Waleed Iqbal, Junaid Qadir, and Ignacio Castro. 2021. Characterising the IETF through the lens of RFC deployment. In *Proceedings of the 21st ACM Internet Measurement Conference* (Virtual Event) *(IMC '21)*. Association for Computing Machinery, New York, NY, USA, 137–149. https://doi.org/10.1145/3487552.3487821

[2] Mehdi Nikkhah, Aman Mangal, Constantine Dovrolis, and Roch Guerin. 2017. A Statistical Exploration of Protocol Adoption. *IEEE/ACM Trans. Netw.* 25, 5 (Oct. 2017), 2858–2871. https://doi.org/10.1109/TNET.2017.2711642